

Topic 4: Introductory Organic Chemistry and Alkanes

Related topics in Units 2, 4 and 5 will assume knowledge of this material.

4A: Introduction

Students will be assessed on their ability to:

4.1	understand the difference between hazard and risk
4.2	understand the hazards associated with organic compounds and why it is necessary to carry out risk assessments when dealing with potentially hazardous materials
4.3	be able to suggest ways in which risks can be reduced and reactions carried out safely, for example: i working on a smaller scale ii taking precautions specific to the hazard iii using an alternative method that involves less hazardous substances
4.4	understand the concepts of homologous series and functional group
4.5	be able to apply the rules of International Union of Pure and Applied Chemistry (IUPAC) nomenclature to: i name compounds relevant to this specification ii draw these compounds, as they are encountered in the specification, using structural, displayed and skeletal formulae <i>Students will be expected to know prefixes for compounds up to C₁₀</i>
4.6	be able to classify reactions as addition, substitution, oxidation, reduction or polymerisation
4.7	understand that bond breaking can be: i homolytic, to produce free radicals ii heterolytic, to produce ions
4.8	know definitions of the terms 'free radical' and 'electrophile'

4B: Alkanes

Students will be assessed on their ability to:

4.9	know the general formula of alkanes and cycloalkanes, and understand that they are hydrocarbons (compounds of carbon and hydrogen only) which are saturated (contain single bonds only)
4.10	understand the term 'structural isomerism' and be able to draw the structural isomers of organic molecules, given their molecular formula
4.11	be able to draw and name the structural isomers of alkanes and cycloalkanes with up to six carbon atoms
4.12	know that alkanes are used as fuels and obtained from the fractional distillation, cracking and reforming of crude oil, and be able to write equations for these reactions
4.13	know that pollutants, including carbon monoxide, oxides of nitrogen and sulfur, carbon particulates and unburned hydrocarbons, are emitted during the combustion of alkane fuels
4.14	understand the problems arising from pollutants from the combustion of alkane fuels, limited to the toxicity of carbon monoxide and why it is toxic, and the acidity of oxides of nitrogen and sulfur
4.15	be able to discuss the reasons for developing alternative fuels in terms of sustainability and reducing emissions, including the emission of CO ₂ and its relationship to climate change
4.16	be able to apply the concept of carbon neutrality to different fuels, such as petrol, bioethanol and hydrogen
4.17	understand the reactions of alkanes with: i oxygen in the air (combustion) ii halogens
4.18	understand the mechanism of the free radical substitution reaction between an alkane and a halogen: i using free radicals, which are species with an unpaired electron, represented by a single dot ii showing the initiation step of the mechanism, with curly half-arrows for free radical formation iii showing the propagation and termination steps of the mechanism iv having limited use in synthesis because of further substitution reactions
	Further suggested practical Cracking alkanes by thermal decomposition, including liquid paraffin using aluminium oxide as a catalyst